Details on addressing incomplete soil sample data
Supporting document IndigoCarbon_US-1_2023_0063

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1 Executive summary

This document fulfills the requirements of Supporting Document IndigoCarbon_US-1_2023_0062 (2023). Based on the results presented below, we have concluded that there is minimal potential for bias due to missing data. Therefore, we are not post-stratifying and are only submitting an unadjusted analysis of reversible emissions reduction. While Supporting Document IndigoCarbon_US-1_2023_0062 (2023) also allows for the use of post-stratification to reduce sampling variance, we have chosen not to pursue this strategy.

Note that in the unadjusted analysis, missing data likely results in increased variance for estimated remissions reduction, and thus a larger uncertainty deduction. Consequently, the unadjusted analysis is automatically penalized for missing data by way of an increased uncertainty deduction. The present document is focused on the potential for missing data to cause bias, and whether any actions are required to account for potential concerns beyond the automatic increase in the uncertainty deduction.

2 Overview

As discussed in Supporting Document IndigoCarbon_US-1_2023_0062 (2023), incomplete (missing) soil sample data have the potential to bias estimates of emissions reduction. Supporting Document IndigoCarbon_US-1_2023_0062 (2023) lays out a series of steps for addressing incomplete soil sample data, and Figure 1 highlights those steps relevant to the current reporting period. The present document presents details for each step shown in Figure 1 that is applicable to the current reporting period.

As described in Supporting Document IndigoCarbon_US-1_2023_0068 (2023), the project has strata referred to as Analysis Strata A, B, C, D, E, F, and G. Reversible emissions reduction were quantified separately for each analysis stratum (with the exception that model variance, and thus the uncertainty deduction, was calculated across all analysis strata; see Supporting Document IndigoCarbon_US-1_2023_0062 (2023) Section 6.2.3 for details). Consequently, the assessments in this document are also specific to each analysis stratum.

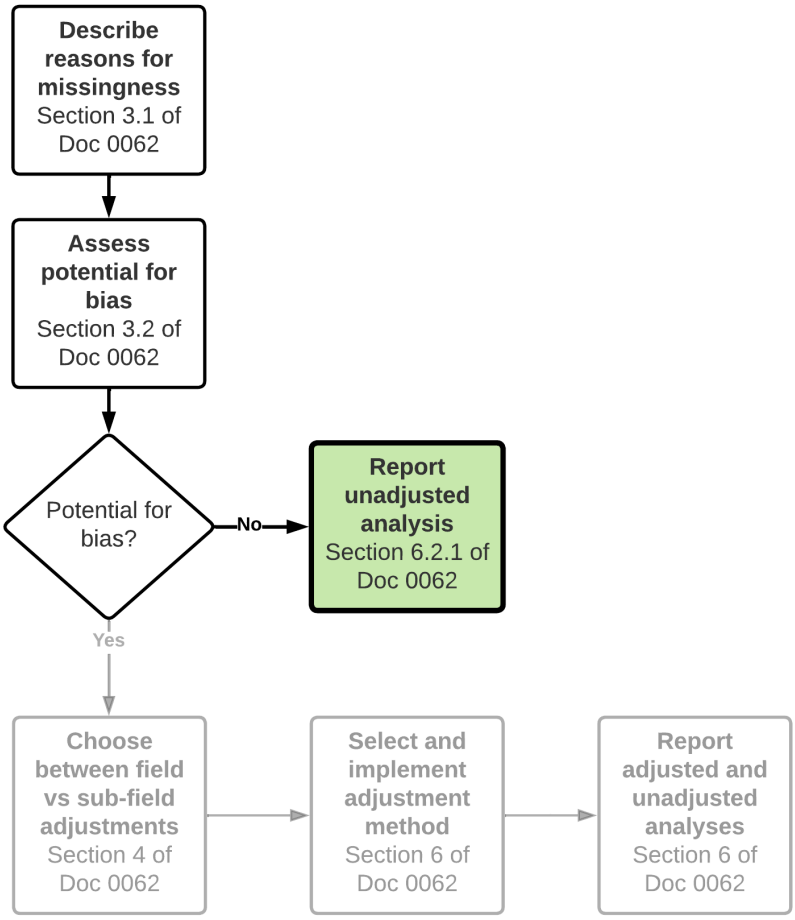


Figure 1: Overall steps and decisions in addressing incomplete soil samples as implemented for this reporting period. Section numbers refer to Supporting Document `IndigoCarbon_US-1_2023_0062` (2023). Grayed out boxes are not relevant for this reporting period; **the three bottom boxes related to adjusted analyses are out-of-scope for the current document.**

3 Reasons for missingness

Sample points that were selected by a sampling plan and intended to be used to quantify reversible emission reductions can be missing due to (1) soil sampling errors, i.e. unable to reliably measure initial percent soil organic carbon (%SOC), (2) biogeochemical model run failures, or (3) quality check failures after model runs. Table 1 tabulates the number of points that are missing from Indigo's analysis due to each of these reasons.

*Modeled changes in SOC for project and baseline scenarios, as well as emissions reduction, were checked to ensure they were less than 5,000 g/m^2 in magnitude. If modeled changes or emissions reduction were larger than 5,000 g/m^2 in magnitude at a given point, that point was discarded. See Supporting Document`IndigoCarbon_US-1_2023_0075` (2023) for details.

Table 2 provides more detailed tabulations on soil sampling errors (missingness that occurs prior to running the model), including misssing %SOC, bulk density, and composite texture and pH samples.

Table 1: Reasons why reversible emissions reduction could not be quantified at points selected by sampling plans

Analysis Stratum	Reason for missingness	Count (% of total planned sample points within analysis stratum)
A	Missing prior to model run (see Table 2)	392 (6.05% of 6475)
	Failed during model run	58 (0.90% of 6475)
	Did not pass quality check after model run*	0 (0.00% of 6475)
	All reasons	450 (6.95% of 6475)
B	Missing prior to model run (see Table 2)	2,596 (27.38% of 9483)
	Failed during model run	113 (1.19% of 9483)
	Did not pass quality check after model run*	0 (0.00% of 9483)
	All reasons	2,709 (28.57% of 9483)
C	Missing prior to model run (see Table 2)	719 (48.88% of 1471)
	Failed during model run	0 (0.00% of 1471)
	Did not pass quality check after model run*	0 (0.00% of 1471)
	All reasons	719 (48.88% of 1471)
D	Missing prior to model run (see Table 2)	327 (15.74% of 2078)
	Failed during model run	79 (3.80% of 2078)
	Did not pass quality check after model run*	0 (0.00% of 2078)
	All reasons	406 (19.54% of 2078)
E	Missing prior to model run (see Table 2)	1,309 (18.45% of 7094)
	Failed during model run	118 (1.66% of 7094)
	Did not pass quality check after model run*	0 (0.00% of 7094)
	All reasons	1,427 (20.12% of 7094)
F	Missing prior to model run (see Table 2)	2,343 (27.63% of 8481)
	Failed during model run	353 (4.16% of 8481)
	Did not pass quality check after model run*	0 (0.00% of 8481)
	All reasons	2,696 (31.79% of 8481)
G	Missing prior to model run (see Table 2)	853 (47.84% of 1783)
	Failed during model run	5 (0.28% of 1783)
	Did not pass quality check after model run*	0 (0.00% of 1783)
	All reasons	858 (48.12% of 1783)
Total	All reasons	9265 (25.13% of 36865)

Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
A	Call plan opened but field not yet sampled	0 (0%)	0 (0%)	15 (2.53% of 593)	15 (2.53% of 593)
	Did not pass quality check prior to model run†	264 (4.12% of 6,475)	96 (4.50% of 2,154)	2 (0.34% of 593)	2 (0.34% of 593)
	Field boundary expanded after randomization‡	69 (1.07% of 6,475)	23 (3.20% of 2,154)	0 (0%)	0 (0%)
	Point outside of boundary	4 (0.06% of 6,475)	0 (0%)	0 (0%)	0 (0%)
	Unknown	55 (0.86% of 6,475)	31 (1.45% of 2,154)	430 (72.51% of 593)	431 (72.68% of 593)
	All reasons	392 (6.11% of 6,475)	150 (9.16% of 2,154)	447 (75.38% of 593)	448 (75.55% of 593)
B	Call plan opened but field not yet sampled	74 (0.99% of 9,483)	26 (1.07% of 3,115)	21 (3.95% of 531)	21 (3.95% of 531)
	Did not pass quality check prior to model run†	395 (5.31% of 9,483)	134 (5.51% of 3,115)	15 (2.82% of 531)	15 (2.82% of 531)
	Field boundary expanded after randomization‡	49 (0.65% of 9,483)	16 (2.00% of 3,115)	0 (0%)	0 (0%)
	Hard soil	1 (0.01% of 9,483)	1 (0.04% of 3,115)	0 (0%)	0 (0%)
	Location inaccessible	7 (0.09% of 9,483)	3 (0.12% of 3,115)	2 (0.38% of 531)	2 (0.38% of 531)
	Point in water	2 (0.03% of 9,483)	1 (0.04% of 3,115)	0 (0%)	0 (0%)

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Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
	Point outside of boundary	13 (0.17% of 9,483)	5 (0.21% of 3,115)	0 (0%)	0 (0%)
	Sampling point not assigned to field staff	1,996 (21.05% of 9,483)	665 (64.08% of 3,115)	0 (0%)	0 (0%)
	Unable to sample to full depth	18 (0.24% of 9,483)	4 (0.16% of 3,115)	3 (0.56% of 531)	3 (0.56% of 531)
	Unknown	41 (0.55% of 9,483)	24 (0.99% of 3,115)	319 (60.08% of 531)	319 (60.08% of 531)
	All reasons	2,596 (29.11% of 9,483)	879 (74.21% of 3,115)	360 (67.80% of 531)	360 (67.80% of 531)
C	Call plan opened but field not yet sampled	12 (1.49% of 1,471)	4 (1.49% of 490)	1 (1.96% of 51)	1 (1.96% of 51)
	Did not pass quality check prior to model run[†]	32 (3.99% of 1,471)	7 (2.61% of 490)	0 (0%)	0 (0%)
	Field boundary expanded after randomization[‡]	1 (0.12% of 1,471)	0 (0.37% of 490)	0 (0%)	0 (0%)
	Location inaccessible	2 (0.25% of 1,471)	0 (0%)	0 (0%)	0 (0%)
	Point outside of boundary	4 (0.50% of 1,471)	3 (1.12% of 490)	2 (3.92% of 51)	2 (3.92% of 51)
	Sampling point not assigned to field staff	667 (45.34% of 1,471)	222 (136.12% of 490)	0 (0%)	0 (0%)
	Unable to sample to full depth	1 (0.12% of 1,471)	0 (0%)	0 (0%)	0 (0%)

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Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
	Unknown	0 (0%)	0 (0%)	23 (45.10% of 51)	23 (45.10% of 51)
	All reasons	719 (51.82% of 1,471)	236 (141.72% of 490)	26 (50.98% of 51)	26 (50.98% of 51)
D	Call plan opened but field not yet sampled	24 (1.30% of 2,078)	8 (1.29% of 695)	1 (0.86% of 116)	1 (0.86% of 116)
	Did not pass quality check prior to model run†	60 (3.24% of 2,078)	22 (3.55% of 695)	6 (5.17% of 116)	6 (5.17% of 116)
	Location inaccessible	1 (0.05% of 2,078)	0 (0%)	0 (0%)	0 (0%)
	Point in water	1 (0.05% of 2,078)	1 (0.16% of 695)	0 (0%)	0 (0%)
	Point outside of boundary	3 (0.16% of 2,078)	1 (0.16% of 695)	0 (0%)	0 (0%)
	Sampling point not assigned to field staff	225 (10.83% of 2,078)	75 (32.37% of 695)	0 (0%)	0 (0%)
	Seasonal delay	0 (0%)	0 (0%)	1 (0.86% of 116)	1 (0.86% of 116)
	Unknown	13 (0.70% of 2,078)	5 (0.81% of 695)	73 (62.93% of 116)	73 (62.93% of 116)
	All reasons	327 (16.33% of 2,078)	112 (38.34% of 695)	81 (69.83% of 116)	81 (69.83% of 116)
E	Call plan opened but field not yet sampled	168 (2.66% of 7,094)	56 (2.65% of 2,370)	13 (2.93% of 444)	13 (2.93% of 444)

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Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
	Did not pass quality check prior to model run†	291 (4.60% of 7,094)	124 (5.87% of 2,370)	24 (5.41% of 444)	24 (5.41% of 444)
	Field boundary expanded after randomization‡	17 (0.27% of 7,094)	6 (0.80% of 2,370)	0 (0%)	0 (0%)
	Hard soil	1 (0.02% of 7,094)	0 (0%)	0 (0%)	0 (0%)
	Location inaccessible	10 (0.16% of 7,094)	5 (0.24% of 2,370)	2 (0.45% of 444)	2 (0.45% of 444)
	Point in water	2 (0.03% of 7,094)	0 (0%)	0 (0%)	0 (0%)
	Point outside of boundary	19 (0.30% of 7,094)	9 (0.43% of 2,370)	4 (0.90% of 444)	4 (0.90% of 444)
	Sampling point not assigned to field staff	755 (10.64% of 7,094)	252 (31.86% of 2,370)	0 (0%)	0 (0%)
	Unable to sample to full depth	8 (0.13% of 7,094)	0 (0%)	1 (0.23% of 444)	1 (0.23% of 444)
	Unknown	38 (0.60% of 7,094)	16 (0.76% of 2,370)	84 (18.92% of 444)	84 (18.92% of 444)
	All reasons	1,309 (19.41% of 7,094)	468 (42.60% of 2,370)	128 (28.83% of 444)	128 (28.83% of 444)
F	Call plan opened but field not yet sampled	1,740 (21.78% of 8,481)	580 (21.79% of 2,826)	121 (19.90% of 608)	121 (19.90% of 608)
	Did not pass quality check prior to model run†	33 (0.41% of 8,481)	93 (3.49% of 2,826)	68 (11.18% of 608)	68 (11.18% of 608)

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Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
	Hard soil	3 (0.04% of 8,481)	1 (0.04% of 2,826)	0 (0%)	0 (0%)
	Location inaccessible	9 (0.11% of 8,481)	5 (0.19% of 2,826)	1 (0.16% of 608)	1 (0.16% of 608)
	Point in water	11 (0.14% of 8,481)	4 (0.15% of 2,826)	1 (0.16% of 608)	1 (0.16% of 608)
	Point outside of boundary	34 (0.43% of 8,481)	13 (0.49% of 2,826)	3 (0.49% of 608)	3 (0.49% of 608)
	Sampling point not assigned to field staff	493 (5.81% of 8,481)	164 (17.45% of 2,826)	0 (0%)	0 (0%)
	Unknown	20 (0.25% of 8,481)	8 (0.30% of 2,826)	84 (13.82% of 608)	99 (16.28% of 608)
	All reasons	2,343 (28.97% of 8,481)	868 (43.89% of 2,826)	278 (45.72% of 608)	293 (48.19% of 608)
G	Call plan opened but field not yet sampled	798 (44.76% of 1,783)	266 (44.78% of 594)	64 (45.39% of 141)	64 (45.39% of 141)
	Did not pass quality check prior to model run[†]	26 (1.46% of 1,783)	3 (0.51% of 594)	0 (0%)	0 (0%)
	Hard soil	3 (0.17% of 1,783)	1 (0.17% of 594)	1 (0.71% of 141)	1 (0.71% of 141)
	Location inaccessible	5 (0.28% of 1,783)	1 (0.17% of 594)	1 (0.71% of 141)	1 (0.71% of 141)
	Point outside of boundary	4 (0.22% of 1,783)	0 (0%)	0 (0%)	0 (0%)

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Table 2: Reasons why soil samples are missing prior to running the model (soil sampling errors)

Analysis Stratum	Reason for missingness	Count (% of total planned samples within analysis stratum) %SOC	Bulk density	pH	Texture
	Unknown	17 (0.95% of 1,783)	5 (0.84% of 594)	6 (4.26% of 141)	6 (4.26% of 141)
	All reasons	853 (47.84% of 1,783)	276 (46.46% of 594)	72 (51.06% of 141)	72 (51.06% of 141)

Notes for Table 2:

† A sample could fail the quality check prior to the biogeochemical model run if any of the following occurred:

- The sample QR code was not valid, the sample QR code was not unique, or the sample was soft-deleted[1] in the database (because it failed sample QC)
- The laboratory QR code was not valid, the laboratory QR code was not unique, or the laboratory result was soft-deleted (because it failed laboratory QC)
- The sample was not taken as prescribed
- An incorrect analysis was done
- The shipment was delayed. A shipment is classified as delayed if: (1) there is more than 21 days between when the sample was collected and received by the lab (used if a lab received date is available), or (2) there is more than 49 days between when the sample was collected and when its result was returned from the lab (used if a lab received date is not available)
- The sample depth was not adequate
- The sampling point was moved beyond the standard operating procedure (SOP) limit (see `IndigoCarbon_US-1_2023_0004`, 2023 for details). Note that points missing due to "Protocol not followed (sample collected at wrong location)" covers all reasons related to taking a sample at the wrong location other than moving the point beyond the SOP limit.

‡ In rare cases, a field boundary was changed after randomization (see Supporting Document `IndigoCarbon_US-1_2023_0034` (2023)). We generated random points in the newly added land at the same sampling density (number of planned points per acre) as in the rest of the pre-stratum (note: not analysis stratum but pre-stratum, i.e., the original stratum at the time of randomization; see `IndigoCarbon_US-1_2023_0068`, 2023 for details on that distinction). Consequently, we only generated points on newly added land if the increase in area multiplied by the sampling density was greater than 1, i.e. if we would have expected sampling points to have been randomly selected in the newly added land had the land been in the pre-stratum at the time of randomization.

* We did additional investigations to better understand why soil samples would be missing due to the reasons: "point in pond," "waterway," and "electric power station." Based on the the areal images shown in Appendix A, we suspect that the reasons "point in pond" and "waterway" were likely due to temporary flooding. We also suspect that the "electric power station" may have been entered erroneously.

4 Assessment of potential bias

Tables 3–9 compare the attributes of i) all points in the sample versus ii) points at which emissions reduction were successfully modeled. These comparisons follow Section 3.2 of Supporting Document

[1] When a sample is "soft-deleted," it is flagged as unusable, but is not actually erased or removed from the database.

IndigoCarbon_US-1_2023_0062 (2023). For categorical attributes, counts and percents are shown in the tables (for each attribute, percents sum to 1 within each column). For continuous attributes, the mean and standard deviation are shown. The right-most column shows the standardized mean difference (SMD) (see Equations (1), (2), and (58) in IndigoCarbon_US-1_2023_0062 (2023)).

In Tables 3–9, the total number of points in the sample are shown under "All", and the total number of observed points is shown under "Observed". Some attributes, such as weather and management practices, change over time. To summarize the spatial-temporal data into summaries for each point, we took the temporal mean for continuous variables (averaged over the project period). For binary variables, such as whether a certain practice change was adopted, we summarized the attribute as "True" if it occurred at any time during the project period, and "False" otherwise.

We took the following approach to collect field attribute data at each point:

- Points for which (latitude, longitude) coordinates were generated by Indigo's call plan (all points in Analysis Stratum A, and points in Analysis Strata B, C, D, E, F, and G for which a call plan was opened): Attributes are pulled based on the geographic location of the planned soil samples.

- Points for which (latitude, longitude) coordinates have not yet generated been generated by a call plan (points in Pre-strata B, C, D, E, F, and G in fields that have not yet been assigned to field staff): We randomly generated (latitude, longitude) coordinates in affected fields so that within each field the total number of generated points is equal to the number of points that will be generated by the call plan when the sampling crews are eventually instructed to sample the field.[2] Attributes were then pulled based on the generated coordinates.

- Additional points that might have been generated by Indigo's call plan in land that was added after randomization (see ‡ note above): We randomly generated (latitude, longitude) coordinates in newly added land so that the number points per acre would be the same as that observed in the rest of the pre-stratum. Attributes were then pulled based on the generated coordinates.

We also note that for attributes with more than two categories, the SMD cannot be calculated if there are two or more categories with zero counts for the number of observed samples. If applicable, these SMDs appear as NaN in Tables 3–9 and will not be shown in Figures 2–8.

Please see Appendix B for the definitions of attributes in Tables 3–9, and please see Table 11 for the definitions of crop functional groups.

[2]The actual call plan may generate different coordinates.

Table 3: Pre-Stratum A

		All (6,406)	Observed (6,025)	SMD
average_relative_humidity_perc		58.20 (7.57)	57.95 (7.59)	-0.034
soil_ph		6.74 (0.69)	6.76 (0.69)	0.019
average_precip_cm		0.18 (0.06)	0.18 (0.06)	-0.033
average_srad_mj_m2		0.66 (0.05)	0.66 (0.05)	0.002
tillage_depth_cm		2.06 (3.33)	2.03 (3.35)	-0.007
silt_pct		53.33 (16.98)	53.08 (16.86)	-0.015
average_tmax_c		19.41 (3.38)	19.39 (3.38)	-0.006
clay_pct		24.44 (9.06)	24.56 (8.99)	0.013
sand_pct		22.23 (19.22)	22.36 (19.26)	0.007
percentage_residue_removed		2.67 (11.77)	2.68 (11.58)	0.001
bulk_density_g_per_cm3		1.36 (0.10)	1.36 (0.10)	-0.025
elevation_m		430.38 (256.53)	439.82 (258.77)	0.037
soil_carbon_pct		1.60 (1.97)	1.62 (2.02)	0.008
continuous_deep_till_to_no_till	False	6,406 (100.00%)	6,025 (100%)	< 0.001
cover_crop_introduction	False	5,189 (81.00%)	4,858 (80.6%)	0.009
	True	1,217 (19.00%)	1,167 (19.4%)	
fertilizer_application_delayed	False	6,001 (93.68%)	5,634 (93.5%)	0.007
	True	405 (6.32%)	391 (6.5%)	
fertilizer_method_impact_reduction	False	6,331 (98.83%)	5,951 (98.8%)	0.005
	True	75 (1.17%)	74 (1.2%)	
legume_introduced_in_cover_crop_mix	False	6,345 (99.05%)	5,965 (99.0%)	0.004
	True	61 (0.95%)	60 (1.0%)	
longer_duration_cover_crop	False	5,982 (93.38%)	5,627 (93.4%)	0.001
	True	424 (6.62%)	398 (6.6%)	
new_cash_crops_in_rotation	False	5,510 (86.01%)	5,174 (85.9%)	0.004
	True	896 (13.99%)	851 (14.1%)	
nitrogen_stabilizer_added	False	6,339 (98.95%)	5,963 (99.0%)	0.002
	True	67 (1.05%)	62 (1.0%)	

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Table 3: Pre-Stratum A

		All (6,406)	Observed (6,025)	SMD
no_till_introduced	True	4,183 (65.30%)	3,971 (65.9%)	0.013
	False	2,223 (34.70%)	2,054 (34.1%)	
tillage_delayed	False	6,406 (100.00%)	6,025 (100%)	< 0.001
tillage_impact_reduction	True	5,122 (79.96%)	4,813 (79.9%)	0.002
	False	1,284 (20.04%)	1,212 (20.1%)	
tillage_passes_reduction	True	4,556 (71.12%)	4,317 (71.7%)	0.012
	False	1,850 (28.88%)	1,708 (28.3%)	
corn	True	3,584 (55.95%)	3,370 (55.9%)	< 0.001
	False	2,822 (44.05%)	2,655 (44.1%)	
cotton	False	6,161 (96.18%)	5,781 (96.0%)	0.012
	True	245 (3.82%)	244 (4.0%)	
soybean	False	3,587 (55.99%)	3,365 (55.9%)	0.003
	True	2,819 (44.01%)	2,660 (44.1%)	
wheat	False	4,354 (67.97%)	4,053 (67.3%)	0.015
	True	2,052 (32.03%)	1,972 (32.7%)	
fallow	False	5,249 (81.94%)	4,958 (82.3%)	0.009
	True	1,157 (18.06%)	1,067 (17.7%)	
chisel	False	6,134 (95.75%)	5,755 (95.5%)	0.012
	True	272 (4.25%)	270 (4.5%)	
deep_ripper	False	6,355 (99.20%)	5,974 (99.2%)	0.006
	True	51 (0.80%)	51 (0.8%)	
disk	False	6,303 (98.39%)	5,923 (98.3%)	0.007
	True	103 (1.61%)	102 (1.7%)	
finisher	False	5,900 (92.10%)	5,542 (92.0%)	0.004
	True	506 (7.90%)	483 (8.0%)	
moldboard	False	6,406 (100.00%)	6,025 (100%)	< 0.001
ridge_till	False	6,406 (100.00%)	6,025 (100%)	< 0.001
row_cultivator	False	6,406 (100.00%)	6,025 (100%)	< 0.001

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Table 3: Pre-Stratum A

		All (6,406)	Observed (6,025)	SMD
strip_till	False	5,691 (88.84%)	5,390 (89.5%)	0.02
	True	715 (11.16%)	635 (10.5%)	
vertical_till	False	5,514 (86.08%)	5,199 (86.3%)	0.006
	True	892 (13.92%)	826 (13.7%)	
lrr_code	M	3,773 (58.90%)	3,476 (57.7%)	0.054
	H	1,307 (20.40%)	1,300 (21.6%)	
	F	1,006 (15.70%)	963 (16.0%)	
	P	115 (1.80%)	113 (1.9%)	
	J	91 (1.42%)	88 (1.5%)	
	I	61 (0.95%)	32 (0.5%)	
	R	41 (0.64%)	41 (0.7%)	
	K	12 (0.19%)	12 (0.2%)	

Table 4: Pre-Stratum B

	All (7,438)	Observed (6,774)	SMD
average_relative_humidity_perc	61.74 (6.71)	61.62 (6.68)	-0.018
soil_ph	6.56 (0.64)	6.56 (0.64)	0.009
average_precip_cm	0.20 (0.07)	0.19 (0.06)	-0.027
average_srad_mj_m2	0.66 (0.05)	0.65 (0.05)	-0.023
tillage_depth_cm	2.03 (3.38)	2.01 (3.34)	-0.005
silt_pct	53.43 (14.89)	53.62 (14.82)	0.013
average_tmax_c	18.89 (3.38)	18.82 (3.37)	-0.022
clay_pct	26.84 (8.59)	26.61 (7.97)	-0.027
sand_pct	19.73 (16.98)	19.76 (16.84)	0.002
percentage_residue_removed	1.04 (7.42)	1.04 (7.54)	0.001

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Table 4: Pre-Stratum B

		All (7,438)	Observed (6,774)	SMD
bulk_density_g_per_cm3		1.37 (0.10)	1.37 (0.10)	-0.014
elevation_m		335.25 (170.48)	338.90 (169.79)	0.021
soil_carbon_pct		1.73 (2.11)	1.75 (2.19)	0.008
continuous_deep_till_to_no_till	False	7,414 (99.68%)	6,752 (99.7%)	< 0.001
	True	24 (0.32%)	22 (0.3%)	
cover_crop_introduction	False	5,306 (71.34%)	4,787 (70.7%)	0.015
	True	2,132 (28.66%)	1,987 (29.3%)	
fertilizer_application_delayed	False	6,932 (93.20%)	6,315 (93.2%)	0.001
	True	506 (6.80%)	459 (6.8%)	
fertilizer_method_impact_reduction	False	7,354 (98.87%)	6,695 (98.8%)	0.003
	True	84 (1.13%)	79 (1.2%)	
legume_introduced_in_cover_crop_mix	False	7,292 (98.04%)	6,666 (98.4%)	0.027
	True	146 (1.96%)	108 (1.6%)	
longer_duration_cover_crop	False	6,621 (89.02%)	6,111 (90.2%)	0.038
	True	817 (10.98%)	663 (9.8%)	
new_cash_crops_in_rotation	False	6,709 (90.20%)	6,110 (90.2%)	< 0.001
	True	729 (9.80%)	664 (9.8%)	
nitrogen_stabilizer_added	False	7,390 (99.35%)	6,726 (99.3%)	0.008
	True	48 (0.65%)	48 (0.7%)	
no_till_introduced	True	4,500 (60.50%)	4,049 (59.8%)	0.015
	False	2,938 (39.50%)	2,725 (40.2%)	
tillage_delayed	False	7,414 (99.68%)	6,750 (99.6%)	0.006
	True	24 (0.32%)	24 (0.4%)	
tillage_impact_reduction	True	5,382 (72.36%)	4,846 (71.5%)	0.018
	False	2,056 (27.64%)	1,928 (28.5%)	
tillage_passes_reduction	True	5,082 (68.32%)	4,577 (67.6%)	0.016
	False	2,356 (31.68%)	2,197 (32.4%)	
corn	False	3,855 (51.83%)	3,540 (52.3%)	0.009

Continued on next page

Table 4: Pre-Stratum B

		All (7,438)	Observed (6,774)	SMD
	True	3,583 (48.17%)	3,234 (47.7%)	
cotton	False	7,134 (95.91%)	6,487 (95.8%)	0.008
	True	304 (4.09%)	287 (4.2%)	
soybean	False	3,861 (51.91%)	3,615 (53.4%)	0.029
	True	3,577 (48.09%)	3,159 (46.6%)	
wheat	False	4,204 (56.52%)	3,850 (56.8%)	0.006
	True	3,234 (43.48%)	2,924 (43.2%)	
fallow	False	6,171 (82.97%)	5,604 (82.7%)	0.006
	True	1,267 (17.03%)	1,170 (17.3%)	
chisel	False	7,144 (96.05%)	6,516 (96.2%)	0.007
	True	294 (3.95%)	258 (3.8%)	
deep_ripper	False	7,333 (98.59%)	6,682 (98.6%)	0.005
	True	105 (1.41%)	92 (1.4%)	
disk	False	7,281 (97.89%)	6,630 (97.9%)	0.001
	True	157 (2.11%)	144 (2.1%)	
finisher	False	6,575 (88.40%)	6,014 (88.8%)	0.012
	True	863 (11.60%)	760 (11.2%)	
moldboard	False	7,426 (99.84%)	6,762 (99.8%)	0.004
	True	12 (0.16%)	12 (0.2%)	
ridge_till	False	7,435 (99.96%)	6,771 (100%)	0.002
	True	3 (0.04%)	3 (0.0%)	
row_cultivator	False	7,275 (97.81%)	6,648 (98.1%)	0.023
	True	163 (2.19%)	126 (1.9%)	
strip_till	False	7,094 (95.38%)	6,484 (95.7%)	0.016
	True	344 (4.62%)	290 (4.3%)	
vertical_till	False	6,267 (84.26%)	5,704 (84.2%)	0.001
	True	1,171 (15.74%)	1,070 (15.8%)	
lrr_code	M	4,614 (62.03%)	4,243 (62.6%)	0.065

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Table 4: Pre-Stratum B

		All (7,438)	Observed (6,774)	SMD
	F	998 (13.42%)	915 (13.5%)	
	H	984 (13.23%)	915 (13.5%)	
	L	203 (2.73%)	198 (2.9%)	
	O	191 (2.57%)	130 (1.9%)	
	P	168 (2.26%)	132 (1.9%)	
	N	132 (1.77%)	108 (1.6%)	
	K	72 (0.97%)	69 (1.0%)	
	G	31 (0.42%)	31 (0.5%)	
	R	21 (0.28%)	21 (0.3%)	
	J	12 (0.16%)	0 (0%)	
	T	12 (0.16%)	12 (0.2%)	

Table 5: Pre-Stratum C

	All (803)	Observed (752)	SMD
average_relative_humidity_perc	62.14 (6.41)	62.26 (6.36)	0.018
soil_ph	6.57 (0.51)	6.56 (0.52)	-0.036
average_precip_cm	0.19 (0.05)	0.18 (0.05)	-0.017
average_srad_mj_m2	0.64 (0.02)	0.64 (0.02)	-0.026
tillage_depth_cm	2.87 (3.40)	2.96 (3.43)	0.028
silt_pct	48.00 (14.95)	47.69 (15.14)	-0.021
average_tmax_c	18.30 (1.48)	18.24 (1.51)	-0.038
clay_pct	26.53 (8.10)	26.24 (8.08)	-0.036
sand_pct	25.47 (18.51)	26.07 (18.79)	0.033
percentage_residue_removed	0.00 (0.00)	0.00 (0.00)	NaN
bulk_density_g_per_cm3	1.39 (0.07)	1.39 (0.07)	0.024

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Table 5: Pre-Stratum C

		All (803)	Observed (752)	SMD
elevation_m		326.67 (138.14)	322.34 (135.78)	-0.031
soil_carbon_pct		1.82 (0.85)	1.82 (0.87)	0.009
continuous_deep_till_to_no_till	False	803 (100.00%)	752 (100%)	< 0.001
cover_crop_introduction	False	618 (76.96%)	580 (77.1%)	0.004
	True	185 (23.04%)	172 (22.9%)	
fertilizer_application_delayed	False	779 (97.01%)	729 (96.9%)	0.004
	True	24 (2.99%)	23 (3.1%)	
fertilizer_method_impact_reduction	False	803 (100.00%)	752 (100%)	< 0.001
legume_introduced_in_cover_crop_mix	False	779 (97.01%)	728 (96.8%)	0.012
	True	24 (2.99%)	24 (3.2%)	
longer_duration_cover_crop	False	767 (95.52%)	717 (95.3%)	0.008
	True	36 (4.48%)	35 (4.7%)	
new_cash_crops_in_rotation	False	791 (98.51%)	740 (98.4%)	0.008
	True	12 (1.49%)	12 (1.6%)	
nitrogen_stabilizer_added	False	791 (98.51%)	740 (98.4%)	0.008
	True	12 (1.49%)	12 (1.6%)	
no_till_introduced	False	518 (64.51%)	504 (67.0%)	0.053
	True	285 (35.49%)	248 (33.0%)	
tillage_delayed	False	803 (100.00%)	752 (100%)	< 0.001
tillage_impact_reduction	True	404 (50.31%)	365 (48.5%)	0.035
	False	399 (49.69%)	387 (51.5%)	
tillage_passes_reduction	False	432 (53.80%)	419 (55.7%)	0.039
	True	371 (46.20%)	333 (44.3%)	
corn	True	563 (70.11%)	520 (69.1%)	0.021
	False	240 (29.89%)	232 (30.9%)	
cotton	False	803 (100.00%)	752 (100%)	< 0.001
soybean	True	466 (58.03%)	439 (58.4%)	0.007
	False	337 (41.97%)	313 (41.6%)	

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Table 5: Pre-Stratum C

		All (803)	Observed (752)	SMD
wheat	False	601 (74.84%)	564 (75.0%)	0.004
	True	202 (25.16%)	188 (25.0%)	
fallow	False	772 (96.14%)	721 (95.9%)	0.014
	True	31 (3.86%)	31 (4.1%)	
chisel	False	775 (96.51%)	724 (96.3%)	0.013
	True	28 (3.49%)	28 (3.7%)	
deep_ripper	False	803 (100.00%)	752 (100%)	< 0.001
disk	False	752 (93.65%)	702 (93.4%)	0.012
	True	51 (6.35%)	50 (6.6%)	
finisher	False	695 (86.55%)	645 (85.8%)	0.023
	True	108 (13.45%)	107 (14.2%)	
moldboard	False	803 (100.00%)	752 (100%)	< 0.001
ridge_till	False	803 (100.00%)	752 (100%)	< 0.001
row_cultivator	False	779 (97.01%)	728 (96.8%)	0.012
	True	24 (2.99%)	24 (3.2%)	
strip_till	False	659 (82.07%)	622 (82.7%)	0.017
	True	144 (17.93%)	130 (17.3%)	
vertical_till	False	624 (77.71%)	586 (77.9%)	0.005
	True	179 (22.29%)	166 (22.1%)	
lrr_code	M	443 (55.17%)	408 (54.3%)	0.026
	L	216 (26.90%)	202 (26.9%)	
	F	96 (11.96%)	95 (12.6%)	
	N	36 (4.48%)	35 (4.7%)	
	P	12 (1.49%)	12 (1.6%)	

Table 6: Pre-Stratum D

		All (1,853)	Observed (1,672)	SMD
average_relative_humidity_perc		62.65 (6.80)	62.42 (6.64)	-0.035
soil_ph		6.41 (0.64)	6.40 (0.61)	-0.015
average_precip_cm		0.20 (0.05)	0.19 (0.05)	-0.036
average_srad_mj_m2		0.66 (0.05)	0.65 (0.04)	-0.048
tillage_depth_cm		2.62 (3.48)	2.68 (3.53)	0.018
silt_pct		51.86 (16.66)	52.51 (16.84)	0.039
average_tmax_c		18.95 (2.99)	18.84 (2.82)	-0.038
clay_pct		26.51 (10.14)	25.69 (9.46)	-0.081
sand_pct		21.62 (18.07)	21.80 (18.38)	0.01
percentage_residue_removed		1.46 (10.28)	1.56 (10.59)	0.009
bulk_density_g_per_cm3		1.39 (0.09)	1.40 (0.08)	0.097
elevation_m		318.81 (199.26)	324.00 (202.29)	0.026
soil_carbon_pct		1.71 (1.01)	1.72 (0.98)	0.002
continuous_deep_till_to_no_till	False	1,853 (100.00%)	1,672 (100%)	< 0.001
cover_crop_introduction	False	1,538 (83.00%)	1,388 (83.0%)	< 0.001
	True	315 (17.00%)	284 (17.0%)	
fertilizer_application_delayed	False	1,739 (93.85%)	1,576 (94.3%)	0.017
	True	114 (6.15%)	96 (5.7%)	
fertilizer_method_impact_reduction	False	1,808 (97.57%)	1,631 (97.5%)	0.002
	True	45 (2.43%)	41 (2.5%)	
legume_introduced_in_cover_crop_mix	False	1,769 (95.47%)	1,638 (98.0%)	0.12
	True	84 (4.53%)	34 (2.0%)	
longer_duration_cover_crop	False	1,658 (89.48%)	1,529 (91.4%)	0.064
	True	195 (10.52%)	143 (8.6%)	
new_cash_crops_in_rotation	False	1,645 (88.77%)	1,502 (89.8%)	0.034
	True	208 (11.23%)	170 (10.2%)	
nitrogen_stabilizer_added	False	1,792 (96.71%)	1,626 (97.2%)	0.03
	True	61 (3.29%)	46 (2.8%)	

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Table 6: Pre-Stratum D

		All (1,853)	Observed (1,672)	SMD
no_till_introduced	True	999 (53.91%)	871 (52.1%)	0.036
	False	854 (46.09%)	801 (47.9%)	
tillage_delayed	False	1,853 (100.00%)	1,672 (100%)	< 0.001
tillage_impact_reduction	True	1,359 (73.34%)	1,217 (72.8%)	0.013
	False	494 (26.66%)	455 (27.2%)	
tillage_passes_reduction	True	1,263 (68.16%)	1,123 (67.2%)	0.021
	False	590 (31.84%)	549 (32.8%)	
corn	True	1,007 (54.34%)	927 (55.4%)	0.022
	False	846 (45.66%)	745 (44.6%)	
cotton	False	1,722 (92.93%)	1,587 (94.9%)	0.077
	True	131 (7.07%)	85 (5.1%)	
soybean	True	930 (50.19%)	800 (47.8%)	0.047
	False	923 (49.81%)	872 (52.2%)	
wheat	False	1,272 (68.65%)	1,181 (70.6%)	0.043
	True	581 (31.35%)	491 (29.4%)	
fallow	False	1,642 (88.61%)	1,478 (88.4%)	0.007
	True	211 (11.39%)	194 (11.6%)	
chisel	False	1,829 (98.70%)	1,651 (98.7%)	0.003
	True	24 (1.30%)	21 (1.3%)	
deep_ripper	False	1,663 (89.75%)	1,502 (89.8%)	0.003
	True	190 (10.25%)	170 (10.2%)	
disk	False	1,841 (99.35%)	1,661 (99.3%)	0.001
	True	12 (0.65%)	11 (0.7%)	
finisher	False	1,531 (82.62%)	1,365 (81.6%)	0.026
	True	322 (17.38%)	307 (18.4%)	
moldboard	False	1,852 (99.95%)	1,671 (99.9%)	0.003
	True	1 (0.05%)	1 (0.1%)	
ridge_till	False	1,853 (100.00%)	1,672 (100%)	< 0.001

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Table 6: Pre-Stratum D

		All (1,853)	Observed (1,672)	SMD
row_cultivator	False	1,853 (100.00%)	1,672 (100%)	< 0.001
strip_till	False	1,781 (96.11%)	1,604 (95.9%)	0.009
	True	72 (3.89%)	68 (4.1%)	
vertical_till	False	1,258 (67.89%)	1,146 (68.5%)	0.014
	True	595 (32.11%)	526 (31.5%)	
lrr_code	M	1,352 (72.96%)	1,238 (74.0%)	0.126
	H	188 (10.15%)	180 (10.8%)	
	L	111 (5.99%)	110 (6.6%)	
	P	84 (4.53%)	33 (2.0%)	
	K	70 (3.78%)	65 (3.9%)	
	F	24 (1.30%)	24 (1.4%)	
	J	24 (1.30%)	22 (1.3%)	

Table 7: Pre-Stratum E

	All (6,322)	Observed (5,667)	SMD
average_relative_humidity_perc	60.23 (5.84)	60.17 (5.80)	-0.01
soil_ph	6.37 (0.60)	6.37 (0.60)	-0.01
average_precip_cm	0.19 (0.06)	0.19 (0.06)	-0.009
average_srad_mj_m2	0.65 (0.05)	0.65 (0.05)	0.018
tillage_depth_cm	2.27 (3.08)	2.29 (3.07)	0.005
silt_pct	56.08 (14.34)	56.00 (14.30)	-0.006
average_tmax_c	19.45 (2.87)	19.48 (2.84)	0.013
clay_pct	27.26 (7.44)	27.27 (7.44)	0.001
sand_pct	16.65 (15.57)	16.73 (15.61)	0.005
percentage_residue_removed	0.48 (5.27)	0.50 (5.51)	0.005

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Table 7: Pre-Stratum E

		All (6,322)	Observed (5,667)	SMD
bulk_density_g_per_cm3		1.34 (0.09)	1.34 (0.09)	0.005
elevation_m		345.33 (179.13)	345.75 (177.12)	0.002
soil_carbon_pct		1.60 (0.85)	1.60 (0.85)	-0.002
continuous_deep_till_to_no_till	False	6,322 (100.00%)	5,667 (100%)	< 0.001
cover_crop_introduction	False	4,824 (76.30%)	4,289 (75.7%)	0.015
	True	1,498 (23.70%)	1,378 (24.3%)	
fertilizer_application_delayed	False	5,894 (93.23%)	5,286 (93.3%)	0.002
	True	428 (6.77%)	381 (6.7%)	
fertilizer_method_impact_reduction	False	6,285 (99.41%)	5,632 (99.4%)	0.004
	True	37 (0.59%)	35 (0.6%)	
legume_introduced_in_cover_crop_mix	False	6,221 (98.40%)	5,584 (98.5%)	0.011
	True	101 (1.60%)	83 (1.5%)	
longer_duration_cover_crop	False	5,903 (93.37%)	5,303 (93.6%)	0.008
	True	419 (6.63%)	364 (6.4%)	
new_cash_crops_in_rotation	False	5,720 (90.48%)	5,136 (90.6%)	0.005
	True	602 (9.52%)	531 (9.4%)	
nitrogen_stabilizer_added	False	6,320 (99.97%)	5,665 (100%)	0.002
	True	2 (0.03%)	2 (0.0%)	
no_till_introduced	True	3,665 (57.97%)	3,251 (57.4%)	0.012
	False	2,657 (42.03%)	2,416 (42.6%)	
tillage_delayed	False	6,322 (100.00%)	5,667 (100%)	< 0.001
tillage_impact_reduction	True	5,029 (79.55%)	4,503 (79.5%)	0.002
	False	1,293 (20.45%)	1,164 (20.5%)	
tillage_passes_reduction	True	4,577 (72.40%)	4,080 (72.0%)	0.009
	False	1,745 (27.60%)	1,587 (28.0%)	
corn	True	3,323 (52.56%)	3,041 (53.7%)	0.022
	False	2,999 (47.44%)	2,626 (46.3%)	
cotton	False	6,214 (98.29%)	5,577 (98.4%)	0.009

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Table 7: Pre-Stratum E

		All (6,322)	Observed (5,667)	SMD
	True	108 (1.71%)	90 (1.6%)	
soybean	True	3,413 (53.99%)	3,024 (53.4%)	0.013
	False	2,909 (46.01%)	2,643 (46.6%)	
wheat	False	3,897 (61.64%)	3,428 (60.5%)	0.024
	True	2,425 (38.36%)	2,239 (39.5%)	
fallow	False	5,645 (89.29%)	5,067 (89.4%)	0.004
	True	677 (10.71%)	600 (10.6%)	
chisel	False	6,184 (97.82%)	5,547 (97.9%)	0.004
	True	138 (2.18%)	120 (2.1%)	
deep_ripper	False	6,274 (99.24%)	5,619 (99.2%)	0.01
	True	48 (0.76%)	48 (0.8%)	
disk	False	6,202 (98.10%)	5,561 (98.1%)	0.002
	True	120 (1.90%)	106 (1.9%)	
finisher	False	5,087 (80.47%)	4,566 (80.6%)	0.003
	True	1,235 (19.53%)	1,101 (19.4%)	
moldboard	False	6,322 (100.00%)	5,667 (100%)	< 0.001
ridge_till	False	6,322 (100.00%)	5,667 (100%)	< 0.001
row_cultivator	False	6,310 (99.81%)	5,655 (99.8%)	0.005
	True	12 (0.19%)	12 (0.2%)	
strip_till	False	6,106 (96.58%)	5,458 (96.3%)	0.015
	True	216 (3.42%)	209 (3.7%)	
vertical_till	False	4,787 (75.72%)	4,322 (76.3%)	0.013
	True	1,535 (24.28%)	1,345 (23.7%)	
lrr_code	M	4,373 (69.17%)	3,887 (68.6%)	0.033
	H	1,307 (20.67%)	1,202 (21.2%)	
	F	347 (5.49%)	328 (5.8%)	
	O	144 (2.28%)	120 (2.1%)	
	N	60 (0.95%)	58 (1.0%)	

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Table 7: Pre-Stratum E

		All (6,322)	Observed (5,667)	SMD
	T	36 (0.57%)	23 (0.4%)	
	G	24 (0.38%)	21 (0.4%)	
	K	19 (0.30%)	19 (0.3%)	
	P	12 (0.19%)	9 (0.2%)	

Table 8: Pre-Stratum F

		All (7,988)	Observed (5,785)	SMD
average_relative_humidity_perc		60.86 (8.52)	61.51 (8.50)	0.077
soil_ph		6.50 (0.82)	6.45 (0.85)	-0.053
average_precip_cm		0.22 (0.10)	0.23 (0.10)	0.063
average_srad_mj_m2		0.67 (0.06)	0.67 (0.05)	0.033
tillage_depth_cm		3.50 (5.35)	3.68 (5.38)	0.032
silt_pct		46.93 (15.84)	47.31 (16.26)	0.024
average_tmax_c		19.69 (3.51)	19.78 (3.43)	0.025
clay_pct		29.25 (13.68)	27.84 (12.42)	-0.103
sand_pct		23.81 (19.41)	24.85 (20.29)	0.053
percentage_residue_removed		0.20 (2.77)	0.05 (1.12)	-0.054
bulk_density_g_per_cm3		1.35 (0.14)	1.36 (0.14)	0.064
elevation_m		340.88 (222.89)	327.67 (218.50)	-0.059
soil_carbon_pct		1.77 (2.22)	1.76 (2.49)	-0.005
continuous_deep_till_to_no_till	False	7,964 (99.70%)	5,773 (99.8%)	0.017
	True	24 (0.30%)	12 (0.2%)	
cover_crop_introduction	False	6,972 (87.28%)	5,006 (86.5%)	0.022
	True	1,016 (12.72%)	779 (13.5%)	
fertilizer_application_delayed	False	7,372 (92.29%)	5,339 (92.3%)	< 0.001

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Table 8: Pre-Stratum F

		All (7,988)	Observed (5,785)	SMD
	True	616 (7.71%)	446 (7.7%)	
fertilizer_method_impact_reduction	False	7,964 (99.70%)	5,761 (99.6%)	0.021
	True	24 (0.30%)	24 (0.4%)	
legume_introduced_in_cover_crop_mix	False	7,880 (98.65%)	5,696 (98.5%)	0.016
	True	108 (1.35%)	89 (1.5%)	
longer_duration_cover_crop	False	7,487 (93.73%)	5,430 (93.9%)	0.006
	True	501 (6.27%)	355 (6.1%)	
new_cash_crops_in_rotation	False	6,871 (86.02%)	5,013 (86.7%)	0.018
	True	1,117 (13.98%)	772 (13.3%)	
nitrogen_stabilizer_added	False	7,267 (90.97%)	5,451 (94.2%)	0.114
	True	721 (9.03%)	334 (5.8%)	
no_till_introduced	True	5,188 (64.95%)	3,652 (63.1%)	0.038
	False	2,800 (35.05%)	2,133 (36.9%)	
tillage_delayed	False	7,988 (100.00%)	5,785 (100%)	< 0.001
tillage_impact_reduction	True	6,835 (85.57%)	4,933 (85.3%)	0.008
	False	1,153 (14.43%)	852 (14.7%)	
tillage_passes_reduction	True	6,512 (81.52%)	4,599 (79.5%)	0.052
	False	1,476 (18.48%)	1,186 (20.5%)	
corn	False	4,377 (54.79%)	3,184 (55.0%)	0.005
	True	3,611 (45.21%)	2,601 (45.0%)	
cotton	False	7,554 (94.57%)	5,386 (93.1%)	0.065
	True	434 (5.43%)	399 (6.9%)	
soybean	False	4,545 (56.90%)	3,159 (54.6%)	0.046
	True	3,443 (43.10%)	2,626 (45.4%)	
wheat	False	5,405 (67.66%)	4,098 (70.8%)	0.068
	True	2,583 (32.34%)	1,687 (29.2%)	
fallow	False	7,330 (91.76%)	5,423 (93.7%)	0.072
	True	658 (8.24%)	362 (6.3%)	

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Table 8: Pre-Stratum F

		All (7,988)	Observed (5,785)	SMD
chisel	False	7,257 (90.85%)	5,262 (91.0%)	0.004
	True	731 (9.15%)	523 (9.0%)	
deep_ripper	False	7,796 (97.60%)	5,619 (97.1%)	0.03
	True	192 (2.40%)	166 (2.9%)	
disk	False	7,892 (98.80%)	5,702 (98.6%)	0.021
	True	96 (1.20%)	83 (1.4%)	
finisher	False	6,922 (86.65%)	4,947 (85.5%)	0.034
	True	1,066 (13.35%)	838 (14.5%)	
moldboard	False	7,988 (100.00%)	5,785 (100%)	< 0.001
ridge_till	False	7,904 (98.95%)	5,716 (98.8%)	0.014
	True	84 (1.05%)	69 (1.2%)	
row_cultivator	False	7,940 (99.40%)	5,742 (99.3%)	0.018
	True	48 (0.60%)	43 (0.7%)	
strip_till	False	7,414 (92.81%)	5,322 (92.0%)	0.032
	True	574 (7.19%)	463 (8.0%)	
vertical_till	False	7,054 (88.31%)	5,044 (87.2%)	0.035
	True	934 (11.69%)	741 (12.8%)	
lrr_code	M	3,402 (42.59%)	2,679 (46.3%)	0.167
	P	1,133 (14.18%)	987 (17.1%)	
	F	934 (11.69%)	484 (8.4%)	
	G	840 (10.52%)	572 (9.9%)	
	O	768 (9.61%)	505 (8.7%)	
	H	588 (7.36%)	359 (6.2%)	
	K	96 (1.20%)	45 (0.8%)	
	L	95 (1.19%)	94 (1.6%)	
	N	84 (1.05%)	49 (0.8%)	
	S	36 (0.45%)	11 (0.2%)	
	J	12 (0.15%)	0 (0%)	

Table 9: Pre-Stratum G

		All (1,783)	Observed (925)	SMD
average_relative_humidity_perc		59.42 (8.58)	59.34 (7.04)	-0.009
soil_ph		6.46 (0.78)	6.37 (0.72)	-0.122
average_precip_cm		0.19 (0.07)	0.19 (0.06)	-0.041
average_srad_mj_m2		0.68 (0.07)	0.67 (0.05)	-0.149
tillage_depth_cm		2.83 (4.14)	2.54 (4.46)	-0.069
silt_pct		52.46 (16.98)	51.81 (16.99)	-0.039
average_tmax_c		20.11 (3.87)	19.91 (2.45)	-0.05
clay_pct		24.91 (9.38)	25.01 (7.07)	0.01
sand_pct		22.62 (19.46)	23.18 (18.54)	0.029
percentage_residue_removed		0.00 (0.00)	0.00 (0.00)	NaN
bulk_density_g_per_cm3		1.36 (0.11)	1.34 (0.12)	-0.175
elevation_m		414.63 (274.95)	383.29 (214.06)	-0.114
soil_carbon_pct		1.55 (0.99)	1.72 (1.13)	0.172
continuous_deep_till_to_no_till	False	1,783 (100.00%)	925 (100%)	< 0.001
cover_crop_introduction	False	1,436 (80.54%)	732 (79.1%)	0.035
	True	347 (19.46%)	193 (20.9%)	
fertilizer_application_delayed	False	1,349 (75.66%)	706 (76.3%)	0.016
	True	434 (24.34%)	219 (23.7%)	
fertilizer_method_impact_reduction	False	1,702 (95.46%)	877 (94.8%)	0.031
	True	81 (4.54%)	48 (5.2%)	
legume_introduced_in_cover_crop_mix	False	1,771 (99.33%)	925 (100%)	0.082
	True	12 (0.67%)	0 (0%)	
longer_duration_cover_crop	False	1,663 (93.27%)	893 (96.5%)	0.131
	True	120 (6.73%)	32 (3.5%)	
new_cash_crops_in_rotation	False	1,595 (89.46%)	846 (91.5%)	0.065
	True	188 (10.54%)	79 (8.5%)	
nitrogen_stabilizer_added	False	1,671 (93.72%)	872 (94.3%)	0.023
	True	112 (6.28%)	53 (5.7%)	

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Table 9: Pre-Stratum G

		All (1,783)	Observed (925)	SMD
no_till_introduced	False	1,004 (56.31%)	458 (49.5%)	0.137
	True	779 (43.69%)	467 (50.5%)	
tillage_delayed	False	1,783 (100.00%)	925 (100%)	< 0.001
tillage_impact_reduction	True	1,115 (62.54%)	611 (66.1%)	0.073
	False	668 (37.46%)	314 (33.9%)	
tillage_passes_reduction	True	1,079 (60.52%)	647 (69.9%)	0.193
	False	704 (39.48%)	278 (30.1%)	
corn	True	1,118 (62.70%)	584 (63.1%)	0.009
	False	665 (37.30%)	341 (36.9%)	
cotton	False	1,752 (98.26%)	894 (96.6%)	0.123
	True	31 (1.74%)	31 (3.4%)	
soybean	False	914 (51.26%)	485 (52.4%)	0.023
	True	869 (48.74%)	440 (47.6%)	
wheat	False	1,292 (72.46%)	689 (74.5%)	0.045
	True	491 (27.54%)	236 (25.5%)	
fallow	False	1,634 (91.64%)	879 (95.0%)	0.122
	True	149 (8.36%)	46 (5.0%)	
chisel	False	1,687 (94.62%)	892 (96.4%)	0.08
	True	96 (5.38%)	33 (3.6%)	
deep_ripper	False	1,723 (96.63%)	889 (96.1%)	0.029
	True	60 (3.37%)	36 (3.9%)	
disk	False	1,699 (95.29%)	901 (97.4%)	0.1
	True	84 (4.71%)	24 (2.6%)	
finisher	False	1,519 (85.19%)	805 (87.0%)	0.052
	True	264 (14.81%)	120 (13.0%)	
moldboard	False	1,783 (100.00%)	925 (100%)	< 0.001
ridge_till	False	1,783 (100.00%)	925 (100%)	< 0.001
row_cultivator	False	1,771 (99.33%)	925 (100%)	0.082

Continued on next page

Table 9: Pre-Stratum G

		All (1,783)	Observed (925)	SMD
	True	12 (0.67%)	0 (0%)	
strip_till	False	1,663 (93.27%)	877 (94.8%)	0.062
	True	120 (6.73%)	48 (5.2%)	
vertical_till	False	1,507 (84.52%)	854 (92.3%)	0.216
	True	276 (15.48%)	71 (7.7%)	
lrr_code	M	803 (45.04%)	556 (60.1%)	NaN
	H	536 (30.06%)	219 (23.7%)	
	P	144 (8.08%)	115 (12.4%)	
	F	132 (7.40%)	12 (1.3%)	
	N	60 (3.37%)	23 (2.5%)	
	O	48 (2.69%)	0 (0%)	
	K	24 (1.35%)	0 (0%)	
	S	12 (0.67%)	0 (0%)	
	G	12 (0.67%)	0 (0%)	
	T	12 (0.67%)	0 (0%)	

Figures 2–8 show the absolute SMDs from Tables 3–9.

5 Discussion and Conclusion

In the majority of cases in which reversible emissions reduction could not be estimated, the cause was logistical and is unlikely to be related to the soil attributes being measured (%SOC, bulk density, pH, and texture). In particular, nearly all missingness occurred prior to model runs (see Table 1), and these missing points are in turn mostly due to sampling logistics (see Table 2).

The SMDs in Tables 3–9 and Figures 2–8 are nearly all less than the 0.1 threshold commonly used in other domains, and all are less than the 0.25 threshold that is sometimes used instead of 0.1 (see Supporting Document `IndigoCarbon_US-1_2023_0062` (2023) for background on the 0.1 and 0.25 thresholds). The exceptions occur in analysis strata F and G, which are newer strata. Sampling is ongoing in stratum G, so the SMDs in that stratum are expected to decline.

While the 0.1 and 0.25 thresholds suggested in Supporting Document `IndigoCarbon_US-1_2023_0062` (2023) are useful starting points for Soil Enrichment Protocol (SEP) projects, it is not clear whether these are the most appropriate thresholds to use for SEP projects. As noted in Supporting Document `IndigoCarbon_US-1_2023_0062` (2023), experience specific to SEP projects, such as the analyses in the present document, will help to determine appropriate thresholds, as well as which attributes are most important for inclusion in Tables 3–9 and Figures 2–8.

Given that: (1) the majority of missing points are missing for reasons that are unlikely to cause bias, and (2) the absolute SMDs are mostly less than 0.25, we have concluded that there is minimal concern for bias due to missing data. Therefore, we have not post-stratified and are only submitting an unadjusted analysis.

In addition to the tabulation of reasons for missingness in Table 2, Supporting Document `IndigoCarbon_US-1_2023_0062` (2023) requests data on: (1) the date at which the problem occurred or was reported, and (2) a 1-2 sentence description of the problem. In compiling this report, we thought that this additional information would not be helpful, and we plan to request a revision to Supporting Document `IndigoCarbon_US-1_2023_0062` (2023) to remove these requirements for future reporting periods. However, if helpful, we could provide dates for all missing soil samples except those for which a field staff has not yet been assigned, or for which a field staff has been assigned but has not yet visited the field. We could also provide slightly more detailed reasons for some but not all missing soil samples. (As Indigo's technology developed, more tools were created to record the reason why a sample was not taken, so older sampling has sparser information about reasons for missingness.)

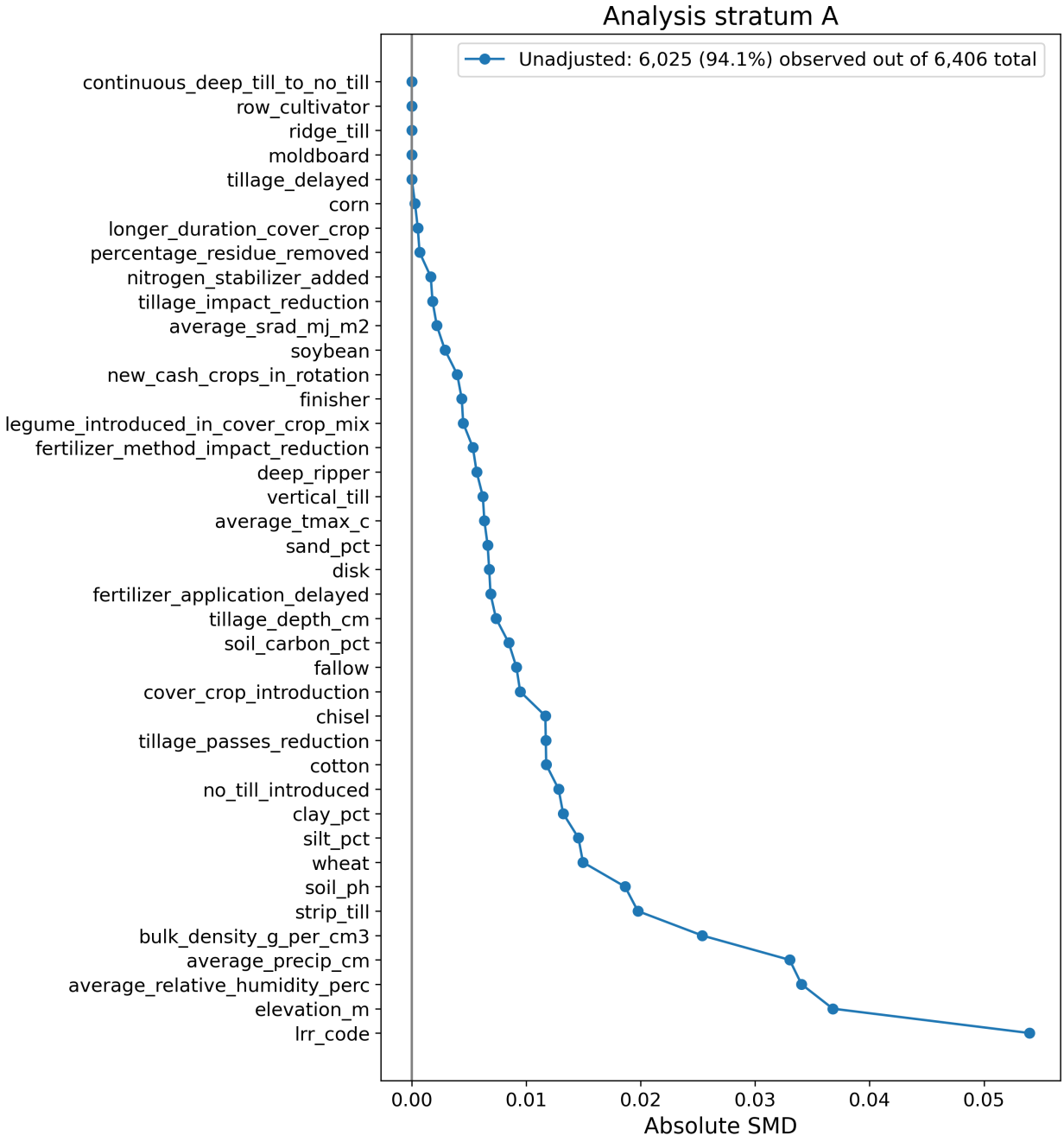


Figure 2: Absolute standardized mean differences (SMDs) for Analysis Stratum A

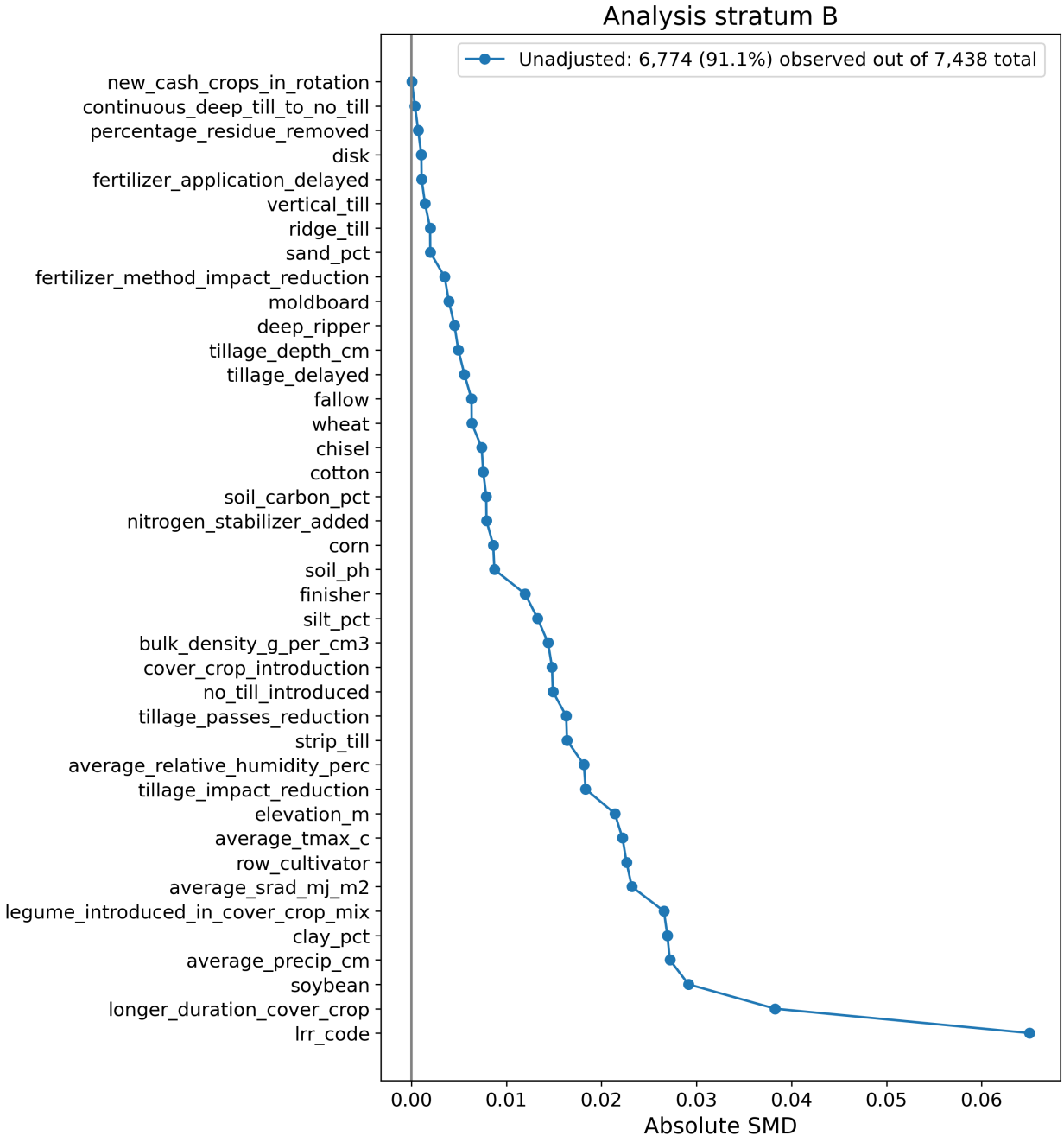


Figure 3: Absolute standardized mean differences (SMDs) for Analysis Stratum B

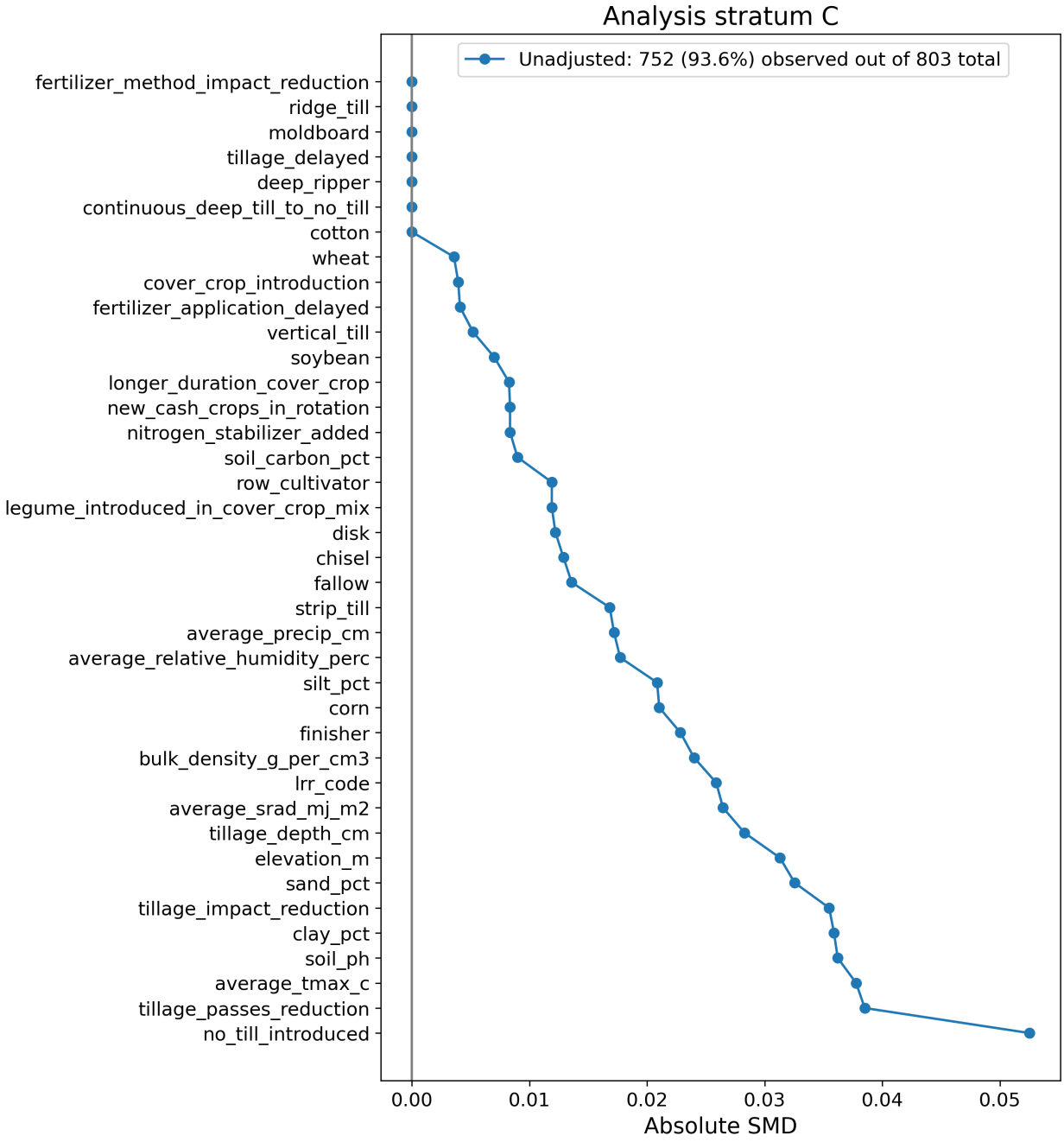


Figure 4: Absolute standardized mean differences (SMDs) for Analysis Stratum C

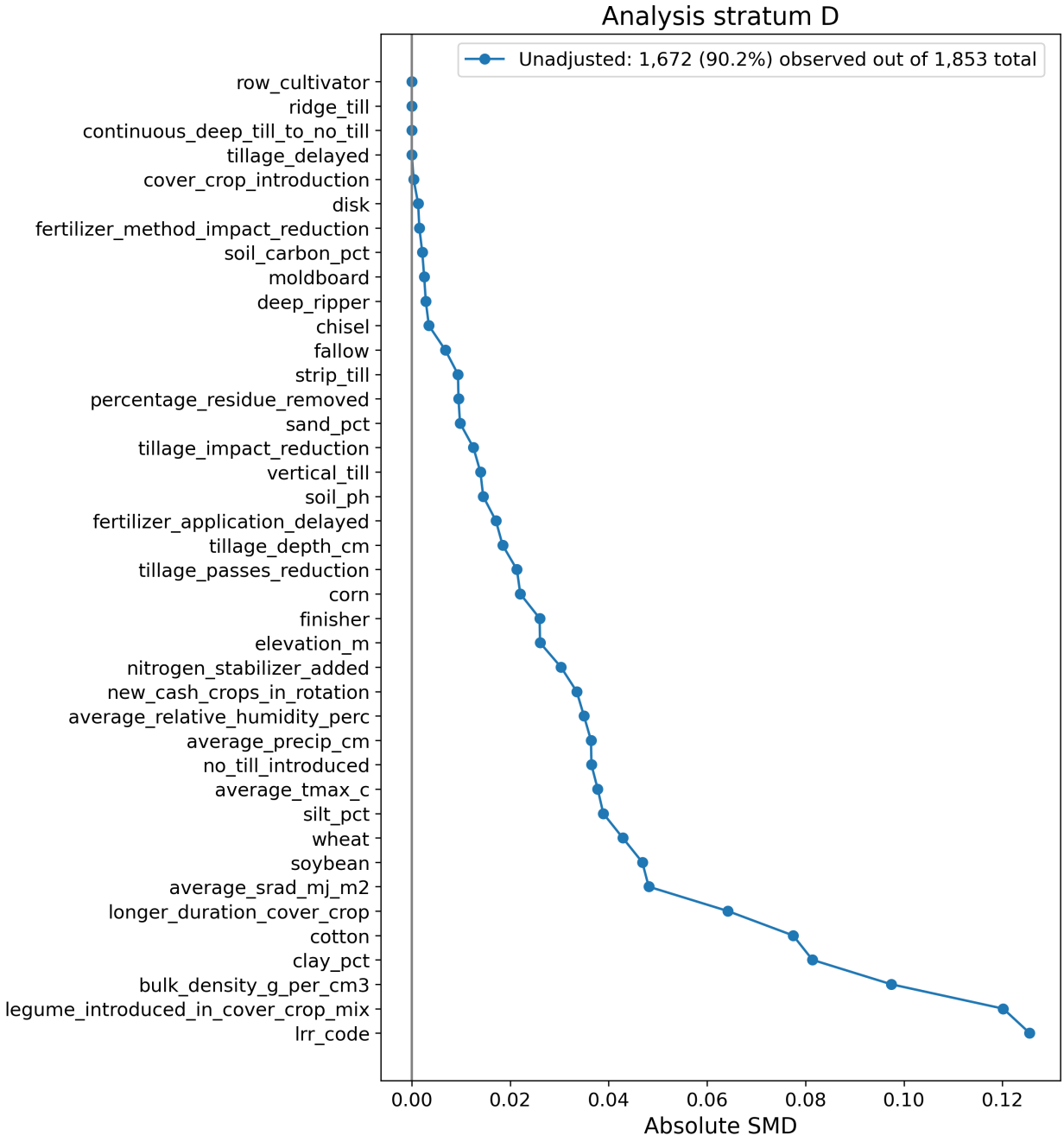


Figure 5: Absolute standardized mean differences (SMDs) for Analysis Stratum D

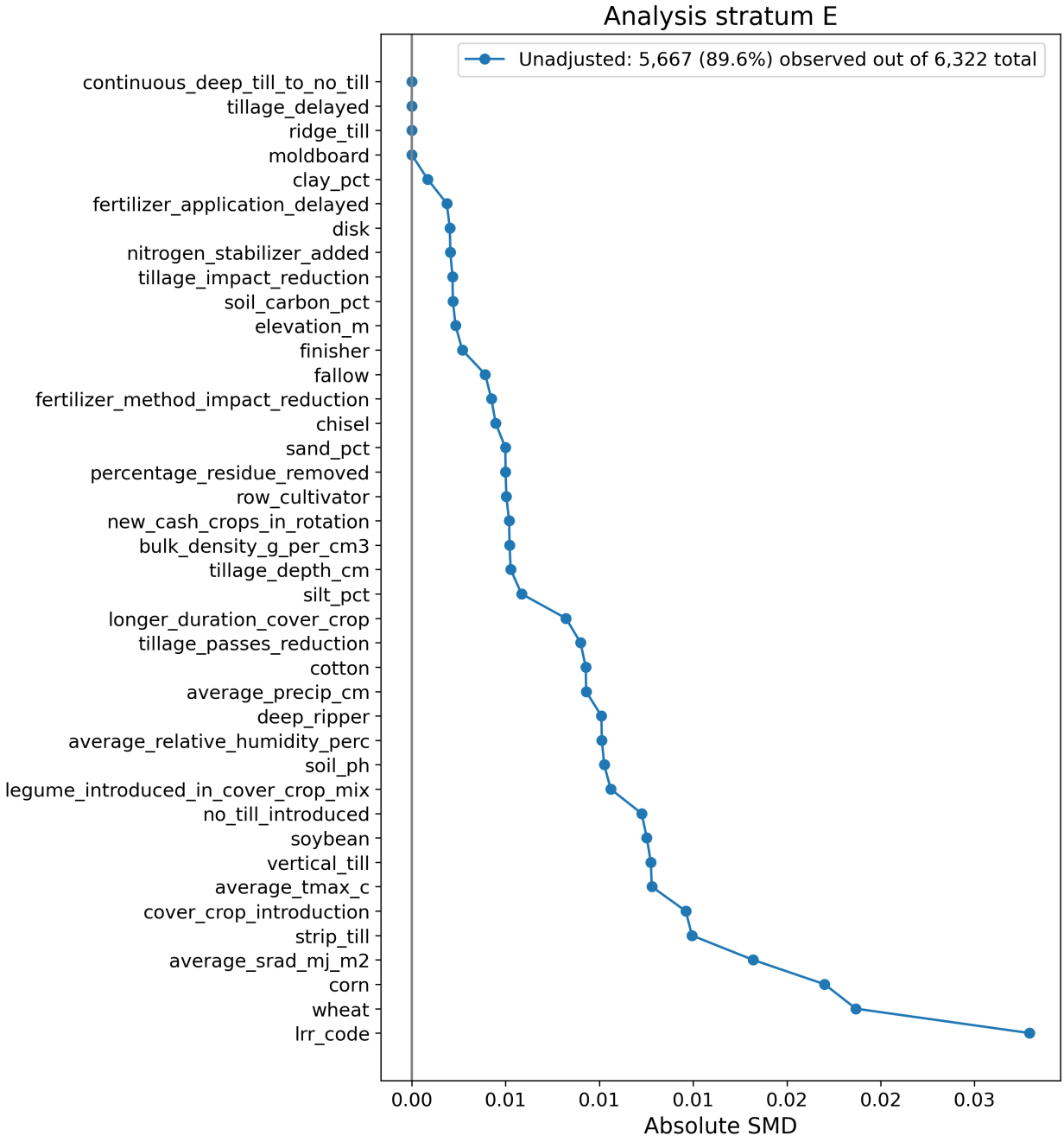


Figure 6: Absolute standardized mean differences (SMDs) for Analysis Stratum E

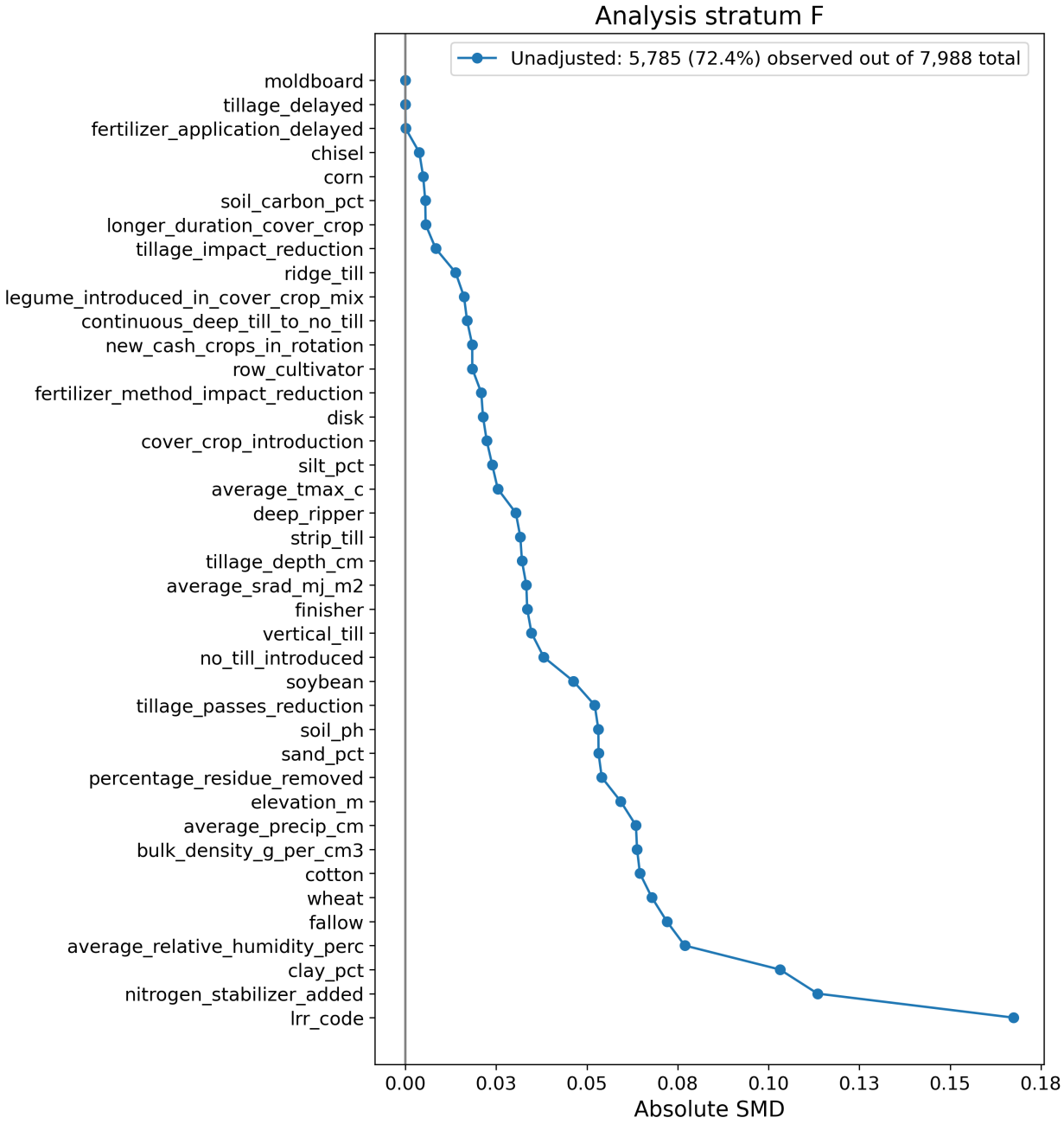


Figure 7: Absolute standardized mean differences (SMDs) for Analysis Stratum F

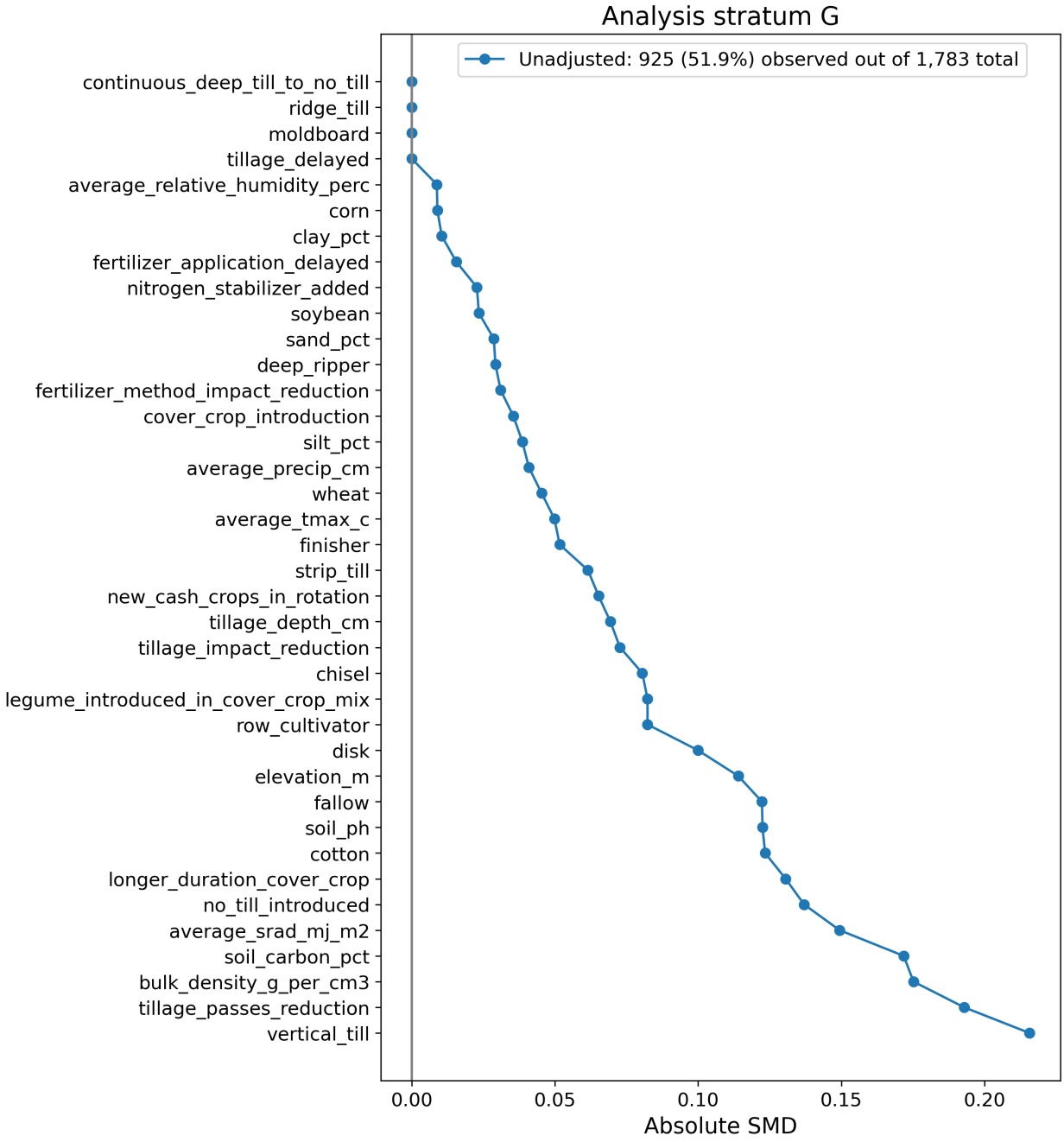


Figure 8: Absolute standardized mean differences (SMDs) for Analysis Stratum G

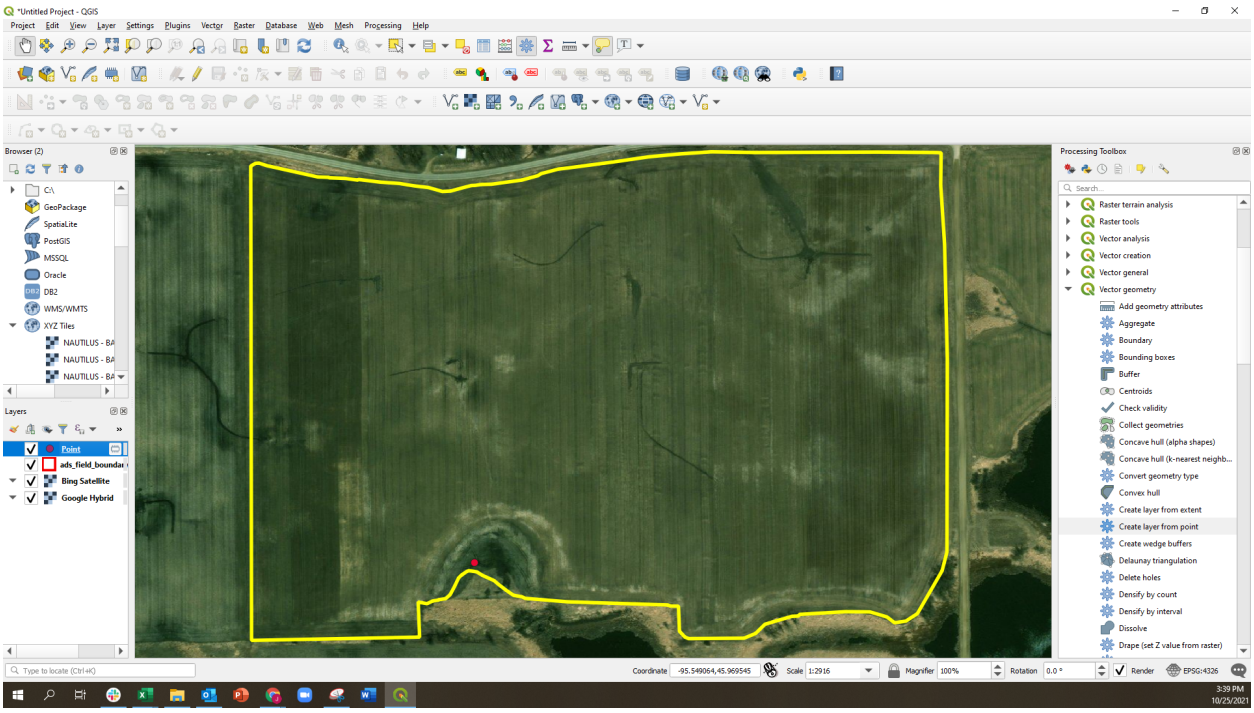


Figure 9: Sample point reportedly missing due to "point in pond" (field_id 31ZQeuXphLM)

A Areal images of selected missing points

Figures 9–10 show the field outline (yellow line) and missing soil sample (red dot) for selected points at which soil samples were planned but are missing.

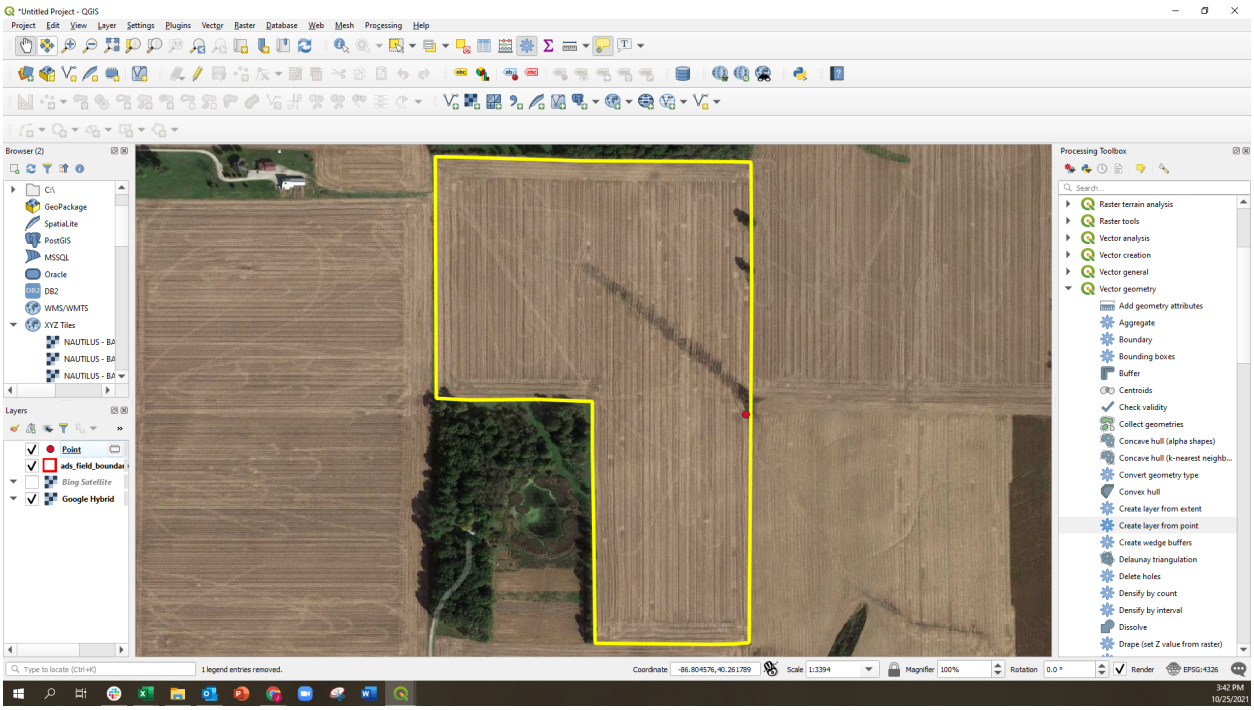


Figure 10: Sample point reportedly missing due to “waterway” (field_id BdoyArR1w8Q)

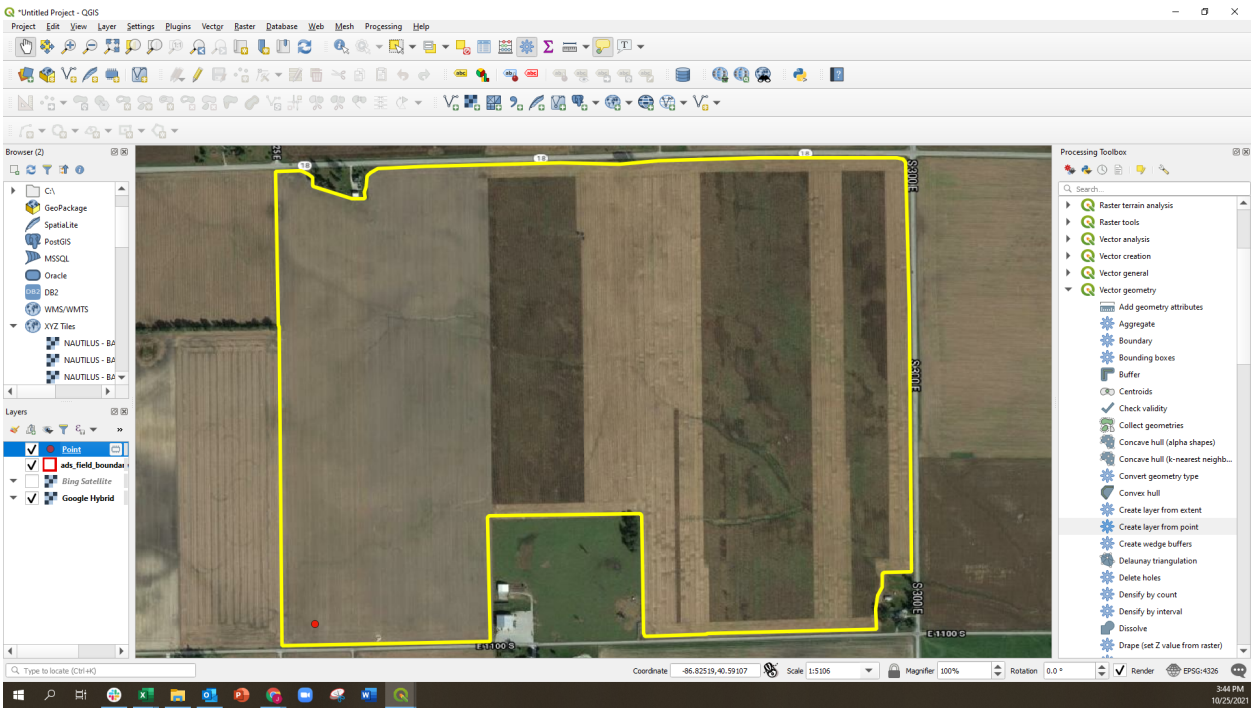


Figure 11: Sample point reportedly missing due to "electric power station" (field_id mEN3RFwhVsk)

B Attribute Definitions

Table 10 shows the definitions and source for each of the attributes shown in Tables 3–9 and Figures 2–8.

Table 10: Attribute definitions

Attribute	Definition	Source
lrr_code	Land resource region	NRCS
soil_carbon_pct	Soil carbon percent (units: percent); temporal mean	SSURGO
bulk_density_g_per_cm3	Bulk density (units: grams / cubic cm); temporal mean	SSURGO
sand_pct	Sand (units: percent); temporal mean	SSURGO
silt_pct	Silt (units: percent); temporal mean	SSURGO
clay_pct	Clay (units: percent); temporal mean	SSURGO
soil_ph	pH; temporal mean	SSURGO
elevation_m	Elevation (units: meters); temporal mean	NLDAS
average_tmin	Daily minimum temperature (units: deg C); temporal mean	NLDAS
average_tmax_c	Daily maximum temperature (units: deg C); temporal mean	NLDAS
average_precip_cm	Daily precipitation (units: cm); temporal mean	NLDAS
average_relative_humidity_perc	Daily average relative humidity (units: percent); temporal mean	NLDAS
average_srad_mj_m2	Daily average solar radiation (units: MJ / m^2); temporal mean	NLDAS
cover_crop_introduction	True if this practice change was adopted, and false otherwise	Grower
fertilizer_application_delayed	True if this practice change was adopted, and false otherwise	Grower
fertilizer_form_impact_reduction	True if this practice change was adopted, and false otherwise	Grower
fertilizer_method_impact_reduction	True if this practice change was adopted, and false otherwise	Grower
fertilizer_reduction	True if this practice change was adopted, and false otherwise	Grower
longer_duration_cover_crop	True if this practice change was adopted, and false otherwise	Grower
new_cash_crops_in_rotation	True if this practice change was adopted, and false otherwise	Grower
nitrogen_stabilizer_added	True if this practice change was adopted, and false otherwise	Grower
no-till_introduced	True if this practice change was adopted, and false otherwise	Grower
synthetic_to_organic_n	True if this practice change was adopted, and false otherwise	Grower
tillage_delayed	True if this practice change was adopted, and false otherwise	Grower
tillage_impact_reduction	True if this practice change was adopted, and false otherwise	Grower

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Table 10: Attribute definitions

Attribute	Definition	Source
tillage_passes_reduction	True if this practice change was adopted, and false otherwise	Grower
wheat	True if a planted crop falls into the wheat crop functional (see Table 11)	Grower
soybean	True if a planted crop falls into the soybean crop functional (see Table 11)	Grower
corn	True if a planted crop falls into the corn crop functional (see Table 11)	Grower
fallow	True if no crop was planted in that cultivation cycle	Grower
percentage_residue_removed	Percent of residual removed (units: percent)	Grower
till_method	The tillage method	Grower
depth_cm	The tillage depth	Grower

Table 11: Names of crop functional groups

Shorthand name	Name used in database
wheat	wheat-barley-rye-oats-flax-canola-potato-tobaco-sunflower-safflower-sugarbeat-cool-grass-triticale
soybean	soybeans-dry-beans-peas-peanuts-alfafa-clover
corn	corn-millet-sorghum-sudangrass
cotton	cotton

Changes in this supporting document since the last verification

- Stratum G has joined the project and is included in the missingness reasons tables and SMD figures.
- We argue that, similarly to previous strata, stratum G does not introduce bias concerns.

References

[1] Supporting document `IndigoCarbon_US-1_2023_0004`. *Field mapping and navigation.* 2023.

[2] Supporting document `IndigoCarbon_US-1_2023_0034`. *Boundary review workflow and standard operating procedures.* 2023.

[3] Supporting document `IndigoCarbon_US-1_2023_0062`. *Addressing incomplete soil sample data in Soil Enrichment Protocol projects (v4.4.1).* 2023.

[4] Supporting document `IndigoCarbon_US-1_2023_0068`. *Details on the pre-strata.* 2023.

[5] Supporting document `IndigoCarbon_US-1_2023_0075`. *Model output screening, QC, and gap-filling assessment.* 2023.